DESIGN AND DEVELOP A WATER LEVEL DETECTION AND FLOOD ALERT SYSTEM USING ULTRASONIC SENSOR.THE SYSTEM MEASURES WATER LEVELS IN RIVERS, TANKS, OR DAMS AND PROVIDES EARLY WARNING ALERTS DURING OVERFLOW CONDITIONS.

A PROJECT REPORT

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1. Project Overview

The Water Level Detection and Flood Alert System using an Ultrasonic Sensor is a smart monitoring system designed to measure water levels continuously in rivers, tanks, dams, and other water bodies. The ultrasonic sensor measures the distance between the sensor and the water surface without direct contact.

The measured water-level data is processed by a microcontroller and compared with predefined safe and critical water-level limits. When the water level reaches a critical value, the system automatically activates an alert such as a buzzer, LED, or warning message. This provides early warning of possible overflow and helps reduce the risk of flooding, property damage, and loss of life.

The system is simple, low-cost, contactless, and suitable for real-time water-level monitoring and flood prevention applications.

2. Problem Statement

Flooding and water overflow can cause serious damage to homes, agricultural land, roads, industries, and public infrastructure. Conventional water-level monitoring methods often require continuous human observation, which may be difficult during heavy rainfall or emergency conditions.

Therefore, there is a need for an automatic and reliable water-level monitoring system that can continuously detect rising water levels and provide an early warning before the water reaches a dangerous level. The proposed system uses an ultrasonic sensor to measure the water level without direct contact and activates an alert when the water level exceeds the predefined safe limit.

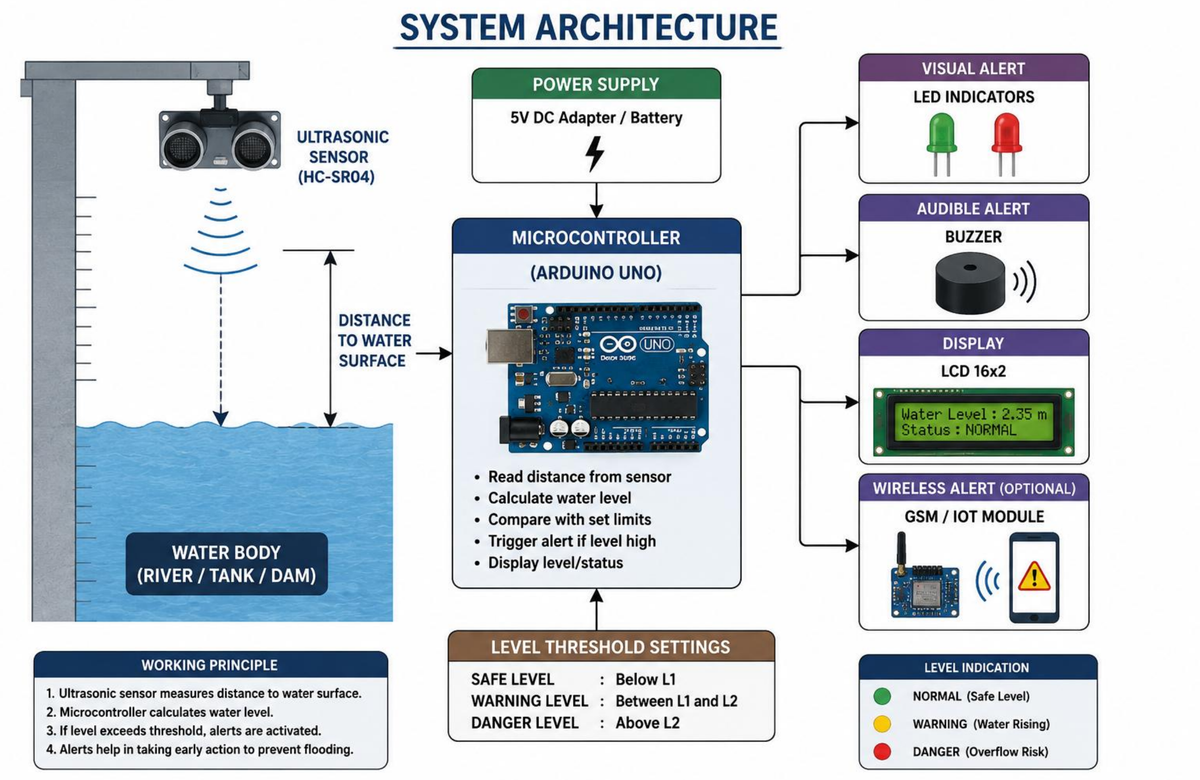
3. Proposed Solution

The proposed system uses an ultrasonic sensor to continuously measure the water level in rivers, tanks, dams, and other water bodies. The sensor sends ultrasonic waves toward the water surface and calculates the distance based on the returning echo.

A microcontroller processes the sensor data and determines the actual water level. The measured level is compared with predefined safe and critical limits. When the water level reaches the critical limit, the system automatically activates a buzzer and warning LED to provide an early flood or overflow alert.

This system provides real-time, contactless, low-cost, and automatic water-level monitoring, reducing the need for continuous human supervision and helping to prevent flood-related damage.

4. System Architecture

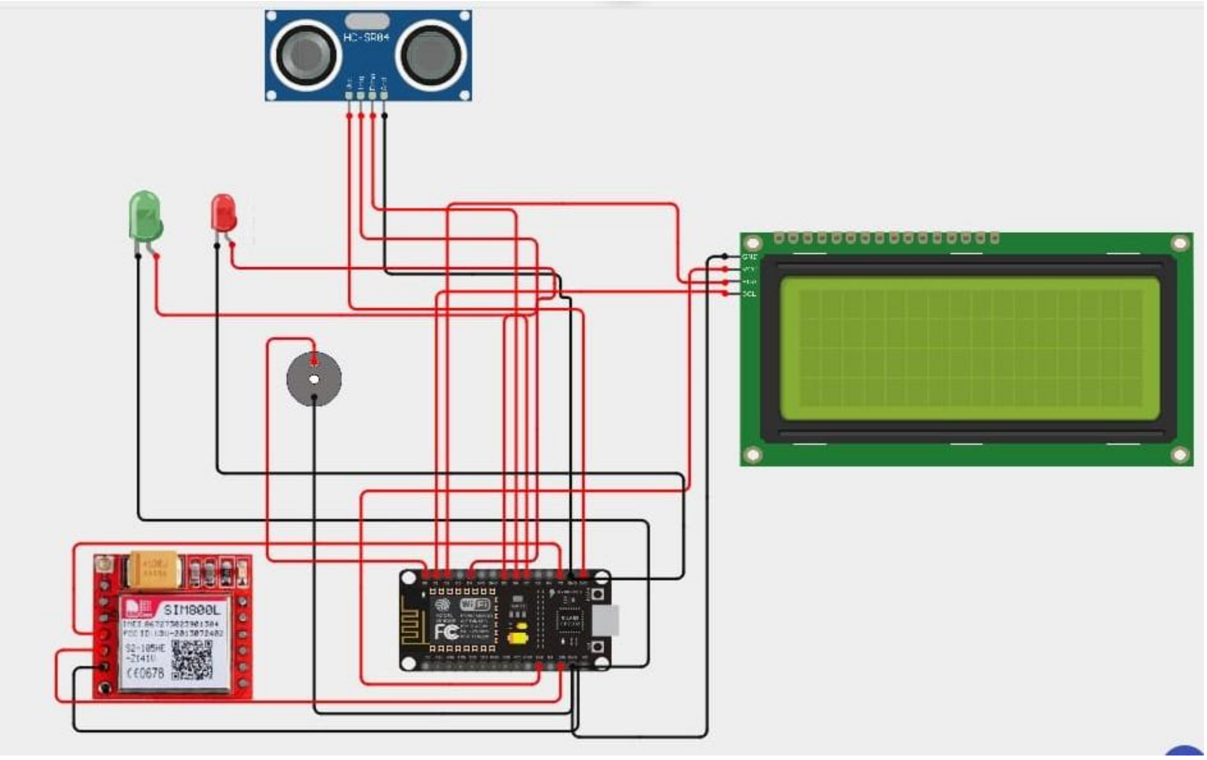


5. Circuit Table

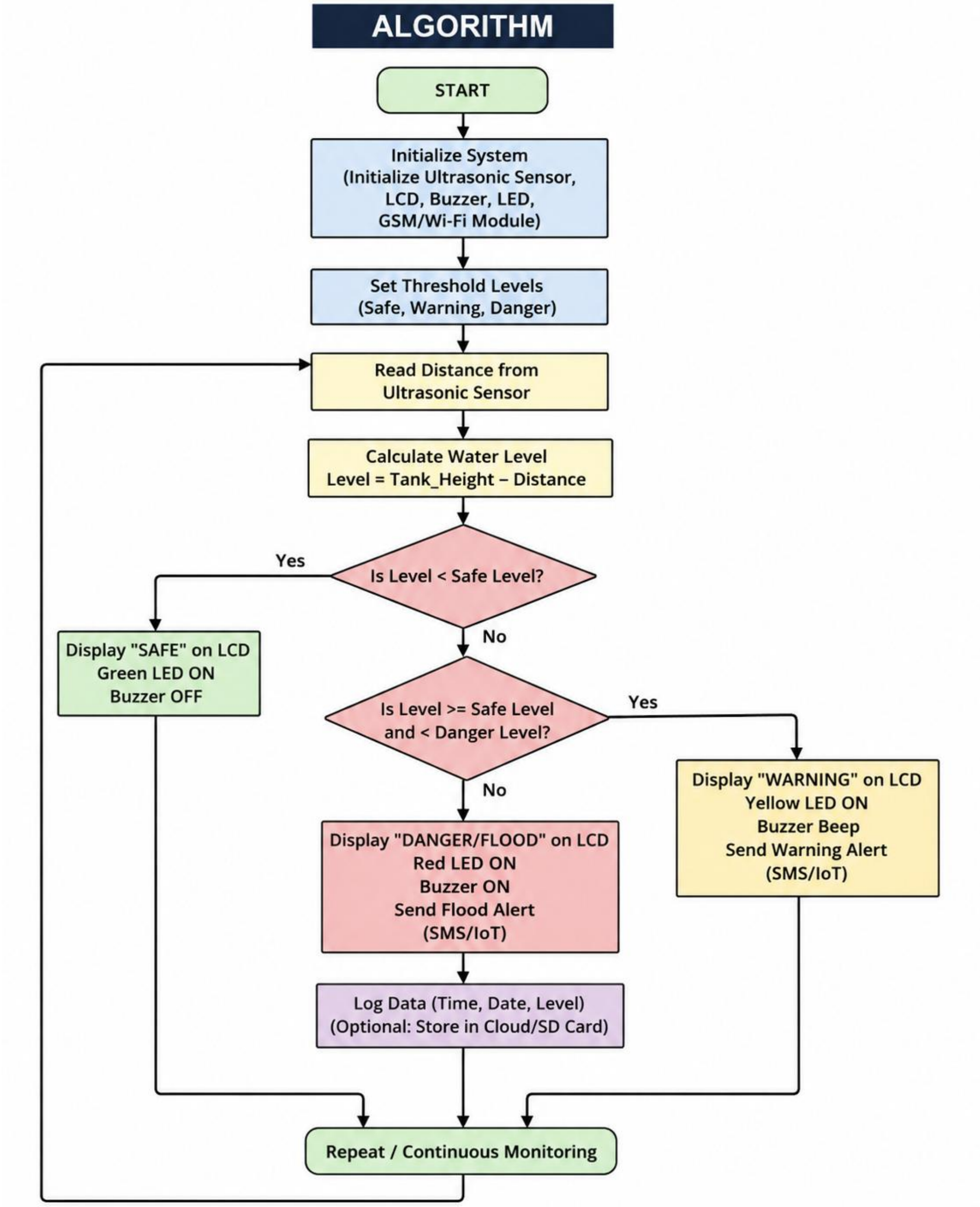
S.No.	Component	Pin/Terminal	Connection
1	Ultrasonic Sensor (HC-SR04)	VCC	Arduino 5V
2	Ultrasonic Sensor	GND	Arduino GND
3	Ultrasonic Sensor	TRIG	Arduino Digital Pin 9
4	Ultrasonic Sensor	ECHO	Arduino Digital Pin 10
5	Buzzer	Positive (+)	Arduino Digital Pin 8
6	Buzzer	Negative (−)	Arduino GND
7	Green LED	Anode (+)	Arduino Digital Pin 6 through 220 Ω resistor
8	Green LED	Cathode (−)	Arduino GND

S.No.	Component	Pin/Terminal	Connection
9	Red LED	Anode (+)	Arduino Digital Pin 7 through 220 Ω resistor
10	Red LED	Cathode (−)	Arduino GND
11	LCD 16×2	VCC	Arduino 5V
12	LCD 16×2	GND	Arduino GND
13	Power Supply	+5V	Arduino 5V
14	Power Supply	GND	Arduino GND

6. Circuit Design



7. Algorithm


ALGORITHM
START
Initialize System
(Initialize Ultrasonic Sensor,
LCD, Buzzer, LED,
GSM/Wi-Fi Module)
Set Threshold Levels
(Safe, Warning, Danger)
Read Distance from
Ultrasonic Sensor
Calculate Water Level
Level = Tank_Height – Distance
Is Level < Safe Level?
Yes
No
Display "SAFE" on LCD
Green LED ON
Buzzer OFF
Is Level >= Safe Level
and < Danger Level?
Yes
No
Display "WARNING" on LCD
Yellow LED ON
Buzzer Beep
Send Warning Alert
(SMS/IoT)
Display "DANGER/FLOOD" on LCD
Red LED ON
Buzzer ON
Send Flood Alert
(SMS/IoT)
Log Data (Time, Date, Level)
(Optional: Store in Cloud/SD Card)
Repeat / Continuous Monitoring

8. Coding

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

// ==================================================
// WIFI
// ==================================================

#define WIFI_SSID     "HITMAN"
#define WIFI_PASSWORD "123456789"

// ==================================================
// THINGSPEAK
// ==================================================

String apiKey = "W9YJD221BI9GOBOH";

const char* server = "http://api.thingspeak.com/update";

// ==================================================
// OLED
// ==================================================

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
```

```
#define OLED_SDA 21
#define OLED_SCL 22

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

// ========================================================
// HC-SR04
// ========================================================

#define TRIG_PIN 5
#define ECHO_PIN 18

#define TANK_HEIGHT 100.0

// ========================================================
// LED + BUZZER
// ========================================================

#define GREEN_LED 25
#define RED_LED 26
#define BUZZER 27

// ========================================================
// GSM
// ========================================================

#define GSM_RX 16
#define GSM_TX 17
```

```
HardwareSerial GSM(2);

String phoneNumber = "+91XXXXXXXXXX";

// =====================================================
// VARIABLES
// =====================================================

float distance;
float waterLevel;
float waterPercentage;

bool alertSent = false;

unsigned long lastThingSpeakUpdate = 0;
const unsigned long thingSpeakInterval = 15000;  // 15 seconds

// =====================================================
// SETUP
// =====================================================

void setup()
{
  Serial.begin(115200);

  // --------------------------------------------------
  // HC-SR04
  // --------------------------------------------------
```

```
pinMode(TRIG_PIN, OUTPUT);
pinMode(ECHO_PIN, INPUT);

// ---------------------------------------------------
// LED + BUZZER
// ---------------------------------------------------

pinMode(GREEN_LED, OUTPUT);
pinMode(RED_LED, OUTPUT);
pinMode(BUZZER, OUTPUT);

digitalWrite(GREEN_LED, LOW);
digitalWrite(RED_LED, LOW);
digitalWrite(BUZZER, LOW);

// ---------------------------------------------------
// OLED
// ---------------------------------------------------

Wire.begin(OLED_SDA, OLED_SCL);

if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
{
  Serial.println("OLED not found!");
  while (1);
}

display.clearDisplay();
display.setTextColor(SSD1306_WHITE);
```

```
display.setTextSize(1);
display.setCursor(20, 20);
display.println("Flood Monitoring");
display.setCursor(35, 35);
display.println("System");

display.display();

// ---------------------------------------------------
// GSM
// ---------------------------------------------------

GSM.begin(9600, SERIAL_8N1, GSM_RX, GSM_TX);

delay(3000);

GSM.println("AT");
delay(1000);

// ---------------------------------------------------
// WIFI
// ---------------------------------------------------

Serial.println();
Serial.println("Connecting to WiFi...");

WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
```

```
  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }

  Serial.println();
  Serial.println("WiFi Connected!");

  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());

  Serial.println("System Started");
}

// ==========================================================
// READ DISTANCE
// ==========================================================

float readDistance()
{
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);

  digitalWrite(TRIG_PIN, LOW);
```

```
  long duration = pulseIn(ECHO_PIN, HIGH, 30000);

  if (duration == 0)
  {
    return -1;
  }

  float distance = duration * 0.0343 / 2.0;

  return distance;
}

// ==========================================================

// SEND SMS

// ==========================================================

void sendSMS(String message)
{
  Serial.println("Sending SMS...");

  GSM.println("AT+CMGF=1");
  delay(1000);

  GSM.print("AT+CMGS=\"");
  GSM.print(phoneNumber);
  GSM.println("\"");

  delay(1000);
```

```
  GSM.print(message);

  delay(500);

  GSM.write(26);

  delay(5000);

  Serial.println("SMS Sent");
}

// ============================================================
// SEND DATA TO THINGSPEAK
// ============================================================

void sendToThingSpeak()
{
  if (WiFi.status() != WL_CONNECTED)
  {
    Serial.println("WiFi disconnected!");
    return;
  }

  HTTPClient http;

  String url = server;

  url += "?api_key=";
  url += apiKey;
```

```
// Field 1 = Water Level
url += "&field1=";
url += String(waterLevel, 1);

// Field 2 = Water Percentage
url += "&field2=";
url += String(waterPercentage, 1);

// Field 3 = Distance
url += "&field3=";
url += String(distance, 1);

// Field 4 = Status
int statusValue;

if (waterPercentage < 60)
{
  statusValue = 0;       // NORMAL
}
else if (waterPercentage < 80)
{
  statusValue = 1;       // WARNING
}
else
{
  statusValue = 2;       // FLOOD
}
```

```
  url += "&field4=";
  url += String(statusValue);

  Serial.println();
  Serial.println("Sending data to ThingSpeak...");

  http.begin(url);

  int httpResponseCode = http.GET();

  if (httpResponseCode > 0)
  {
    Serial.print("ThingSpeak Response: ");
    Serial.println(httpResponseCode);

    Serial.print("Entry ID: ");
    Serial.println(http.getString());
  }
  else
  {
    Serial.print("ThingSpeak Error: ");
    Serial.println(httpResponseCode);
  }

  http.end();
}

// ========================================================
// LOOP
```

```
// =====================================================

void loop()
{
  distance = readDistance();

  if (distance < 0)
  {
    Serial.println("Ultrasonic sensor error");

    display.clearDisplay();
    display.setTextSize(1);
    display.setCursor(15, 25);
    display.println("Sensor Error");
    display.display();

    delay(1000);
    return;
  }

  // Limit distance
  if (distance > TANK_HEIGHT)
  {
    distance = TANK_HEIGHT;
  }

  if (distance < 0)
  {
    distance = 0;
```

```
}

// --------------------------------------------------

// CALCULATE WATER LEVEL

// --------------------------------------------------


waterLevel = TANK_HEIGHT - distance;


waterPercentage = (waterLevel / TANK_HEIGHT) * 100.0;


// --------------------------------------------------

// SERIAL MONITOR

// --------------------------------------------------


Serial.println("----------------------------");


Serial.print("Distance: ");

Serial.print(distance);

Serial.println(" cm");


Serial.print("Water Level: ");

Serial.print(waterLevel);

Serial.println(" cm");


Serial.print("Water Percentage: ");

Serial.print(waterPercentage);

Serial.println(" %");


// --------------------------------------------------
```

```
// NORMAL
// ------------------------------------------------

if (waterPercentage < 60)
{
  digitalWrite(GREEN_LED, HIGH);
  digitalWrite(RED_LED, LOW);
  digitalWrite(BUZZER, LOW);

  alertSent = false;

  Serial.println("Status: NORMAL");
}

// ------------------------------------------------
// WARNING
// ------------------------------------------------

else if (waterPercentage >= 60 && waterPercentage < 80)
{
  digitalWrite(GREEN_LED, HIGH);
  digitalWrite(RED_LED, LOW);
  digitalWrite(BUZZER, LOW);

  Serial.println("Status: WARNING");
}

// ------------------------------------------------
// FLOOD ALERT
```

```
// ----------------------------------------------------

else
{
  digitalWrite(GREEN_LED, LOW);
  digitalWrite(RED_LED, HIGH);

  digitalWrite(BUZZER, HIGH);
  delay(300);
  digitalWrite(BUZZER, LOW);

  Serial.println("Status: FLOOD ALERT!");

  // Send SMS only once
  if (!alertSent)
  {
    String message = "FLOOD ALERT!\n";
    message += "Water Level: ";
    message += String(waterLevel, 1);
    message += " cm\n";
    message += "Water Level: ";
    message += String(waterPercentage, 1);
    message += "%";

    sendSMS(message);

    alertSent = true;
  }
}
```

```
// ---------------------------------------------------
// OLED
// ---------------------------------------------------

display.clearDisplay();

display.setTextSize(1);
display.setCursor(0, 0);
display.println("WATER LEVEL MONITOR");

display.drawLine(0, 10, 127, 10, SSD1306_WHITE);

display.setCursor(0, 18);
display.print("Distance: ");
display.print(distance, 1);
display.println(" cm");

display.setCursor(0, 30);
display.print("Level: ");
display.print(waterLevel, 1);
display.println(" cm");

display.setCursor(0, 42);
display.print("Level: ");
display.print(waterPercentage, 1);
display.println("%");

display.setCursor(0, 54);
```

```
  if (waterPercentage < 60)
  {
    display.println("STATUS: NORMAL");
  }
  else if (waterPercentage < 80)
  {
    display.println("STATUS: WARNING");
  }
  else
  {
    display.println("!!! FLOOD ALERT !!!");
  }

  display.display();

  // --------------------------------------------------
  // THINGSPEAK UPDATE
  // --------------------------------------------------

  if (millis() - lastThingSpeakUpdate >= thingSpeakInterval)
  {
    sendToThingSpeak();

    lastThingSpeakUpdate = millis();
  }

  delay(2000);
}
```

9. System Working

1. Water Level Detection: The HC-SR04 ultrasonic sensor is placed above the water surface and sends ultrasonic waves toward the water.
2. Distance Measurement: The sensor receives the reflected waves and calculates the distance between the sensor and the water surface.
3. Data Processing: The Arduino UNO receives the distance value and calculates the actual water level.
4. Level Comparison: The measured water level is continuously compared with predefined safe, warning, and danger limits.
5. Normal Condition: When the water level is within the safe limit, the green LED remains ON and the system continues monitoring.
6. Warning Condition: When the water level rises above the warning limit, the system activates a warning indication to alert the user.
7. Danger Condition: When the water reaches the critical level, the red LED and buzzer are activated to provide an immediate flood/overflow alert.
8. Continuous Monitoring: The system continuously repeats the measurement process, allowing real-time monitoring and providing early warning before overflow occurs.

10. Bill of Materials

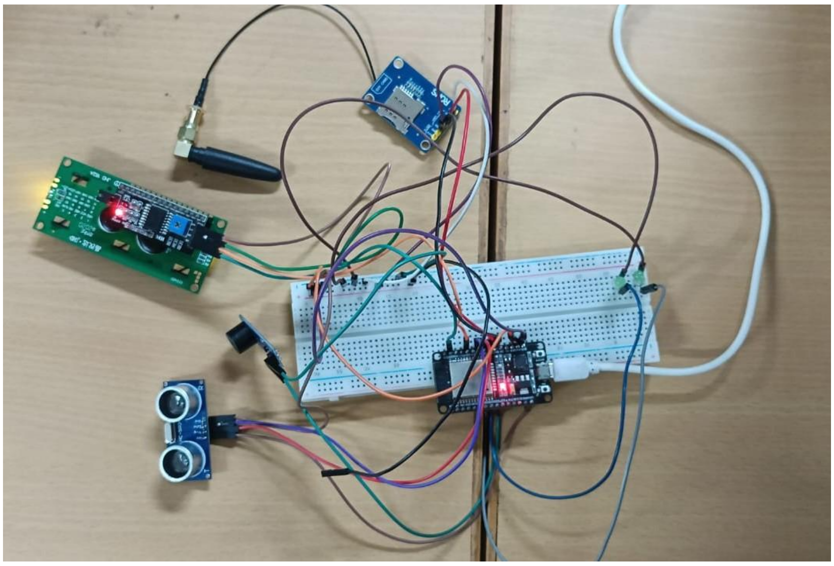
S.No.	Component	Specification	Quantity
1	Arduino UNO	ATmega328P	1
2	Ultrasonic Sensor	HC-SR04	1
3	Buzzer	5V DC	1
4	Green LED	5mm	1
5	Red LED	5mm	1
6	Resistor	220 Ω	2
7	LCD Display	16×2	1
8	Breadboard	Standard size	1
9	Jumper Wires	Male-Male / Male-Female	As required

S.No.	Component	Specification	Quantity
10	Power Supply	5V USB / Adapter	1
11	Connecting Wires	Single-core	As required
12	Water Container/Model	Prototype tank	1

11. Prototype Implementation

The prototype of the Water Level Detection and Flood Alert System is developed using an Arduino UNO, HC-SR04 ultrasonic sensor, LEDs, buzzer, LCD display, breadboard, and jumper wires.

- The ultrasonic sensor is mounted above a water tank to measure the distance between the sensor and the water surface.
- The Arduino UNO is connected to the sensor and programmed to calculate the water level.
- A green LED indicates the safe water level.
- A red LED and buzzer provide an alert when the water level reaches the critical limit.
- The 16×2 LCD displays the measured water level or status.
- Different water levels are tested by gradually filling the tank with water.
- When the water reaches the predefined danger level, the buzzer and red LED automatically turn ON.
- The prototype demonstrates real-time, automatic, and contactless water-level monitoring suitable for flood and overflow warning applications.



12. Results & Performance Analysis

Results:

The developed prototype successfully detects the water level using the HC-SR04 ultrasonic sensor and displays the water-level status. When the water level increases beyond the predefined limit, the red LED and buzzer are activated, providing an early warning of possible overflow.

Performance Analysis:

- The system provides continuous real-time water-level monitoring.
- The ultrasonic sensor measures the water level without direct contact.
- The alert system responds automatically when the critical level is reached.
- The prototype is simple, low-cost, and easy to operate.
- It reduces the need for continuous human monitoring.
- The system can be further improved by adding GSM/IoT technology for remote flood alerts.

Overall Performance:

The prototype demonstrates that the proposed system can effectively monitor rising water levels and provide early flood and overflow warnings, making it suitable for small-scale tanks, dams, rivers, and flood-monitoring applications.